

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITMTB1-44

Group: 2

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Indicate	Yes	No
Plagiarism report attached		

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Question 1

1a)

$$\begin{aligned}
 Rn &= \sum_{i=1}^n f(xi)\Delta x \text{ (Math, 2020)} \\
 Rn &= (2-0)(60) + (5-2)(90) + (7-5)(110) + (8-7)(120) \\
 &= (2)(60) + (3)(90) + (2)(110) + (1)(120) \\
 &= 730 \text{ MB of data was removed at 8 seconds}
 \end{aligned}$$

1b)

$$\begin{aligned}
 &\int_0^8 400e^{-\frac{t^2}{4}} dt \text{ (Wizard, 2021)} \\
 &= 400 \int_0^8 e^{-\frac{t^2}{4}} dt \\
 \text{Let } e^t &= \sum_{n=0}^{\infty} \frac{t^n}{n!} = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \dots \\
 e^{-\frac{t^2}{4}} &= 1 - \frac{t^2}{4} + \frac{\frac{1}{16}t^4}{2!} - \frac{\frac{1}{64}t^6}{3!} + \frac{\frac{1}{256}t^8}{4!} + \dots \\
 400 \int_0^8 e^{-\frac{t^2}{4}} dt &= 400 \int_0^8 1 - \frac{t^2}{4} + \frac{\frac{1}{16}t^4}{2!} - \frac{\frac{1}{64}t^6}{3!} + \frac{\frac{1}{256}t^8}{4!} + \dots dt \\
 &= 400 \left[t - \frac{t^3}{12} + \frac{\frac{1}{16}t^5}{5 \cdot 2!} - \frac{\frac{1}{64}t^7}{7 \cdot 3!} + \frac{\frac{1}{256}t^9}{9 \cdot 4!} \right]_0^8 \\
 &= 400 \left[8 - \frac{8^3}{12} + \frac{\frac{1}{16}(8)^5}{10} - \frac{\frac{1}{64}(8)^7}{42} + \frac{\frac{1}{256}(8)^9}{216} \right]_0^8 \\
 &= 400(1817, 20) = 726880, 85
 \end{aligned}$$

$$100 + 726880, 85 - 730 = 726250, 85 \text{ MB is left in the server at the end of 8 hours}$$

1c)

$$\begin{aligned}
 &\text{Slope1 : (M119, 2013)} \\
 &x = 5 \text{ to } x = 7 \\
 &\frac{120-110}{7-5} = 5
 \end{aligned}$$

Slope2 :

$$\begin{aligned}
 &x = 2 \text{ to } x = 5 \\
 &\frac{110-90}{5-2} = \frac{20}{3}
 \end{aligned}$$

$$R'(5) \approx \frac{5 + \frac{20}{3}}{2} \approx \frac{35}{6}$$

Question 2

2.1a)

The average monthly consumption will need to be calculated over a period of twelve months from $t = 0$ to $t = 12$.

$$Ravg = \frac{1}{b-a} \int_a^b f(x)dx$$

$$= \frac{1}{12-0} \int_0^{12} ((11, 21 - \cos(\frac{\pi t}{6})) \times 10^9) dx$$

$$= \frac{1}{12} \cdot 10^9 \int_0^{12} (11, 21 - \cos(\frac{\pi t}{6})) dx$$

$$\text{Let } I = \int (11, 21 - \cos(\frac{\pi t}{6})) dx$$

$$\text{Let } u = \frac{\pi t}{6} = \frac{\pi}{6} t$$

$$\frac{du}{dt} = \frac{\pi}{6}$$

$$du = \frac{\pi}{6} dt$$

$$I = \int (11, 21 - \frac{\pi}{6} \cos(u)) du$$

$$= 11, 21t - \frac{\pi}{6} \sin(u) + c$$

$$= 11, 21t - \frac{\pi}{6} \sin(\frac{\pi t}{6}) + c$$

$$Ravg = \frac{10^9}{12} [11, 21t - \frac{\pi}{6} \sin(\frac{\pi t}{6})]_0^{12} = \frac{10^9}{12} (\frac{3363}{25} - 0) = 1, 121.10^{10}$$

2.1b)

$$1, 121.10^{10} \times 12 = 1, 34.10^{11} \text{ litres of petrol}$$

2.1c)

$$Ravg = \frac{1}{b-a} \int_a^b f(x) dx$$

$$= \frac{1}{9-3} \int_3^9 ((11, 21 - \cos(\frac{\pi t}{6})) \times 10^9) dx$$

$$= \frac{1}{6} \cdot 10^9 \int_3^9 (11, 21 - \cos(\frac{\pi t}{6})) dx$$

$$\text{Let } I = \int (11, 21 - \cos(\frac{\pi t}{6})) dx$$

$$\text{Let } u = \frac{\pi t}{6} = \frac{\pi}{6} t$$

$$\frac{du}{dt} = \frac{\pi}{6}$$

$$du = \frac{\pi}{6} dt$$

$$I = \int (11, 21 - \frac{\pi}{6} \cos(u)) du$$

$$= 11, 21t - \frac{\pi}{6} \sin(u) + c$$

$$= 11, 21t - \frac{\pi}{6} \sin(\frac{\pi t}{6}) + c$$

$$Ravg = \frac{10^9}{6} [11, 21t - \frac{\pi}{6} \sin(\frac{\pi t}{6})]_3^9 = \frac{10^9}{6} (101, 41 - 33, 11) = 1, 14.10^{10}$$

2.2)

$$\int_{-2}^n (4s + 1) ds = 30$$

$$\frac{4s^2}{2} + s \Big|_{-2}^n = 30$$

$$2s^2 + s \Big|_{-2}^n = 30$$

$$2n^2 + n - (2(-2)^2 - 2) = 30$$

$$2n^2 + n - (6) = 30$$

$$2n^2 + n - 36 = 0$$

$$(2n + 9)(n - 4) = 0$$

$$2n = -9 \text{ or } n = 4$$

$$n \neq -\frac{9}{2}$$

$$2.3a)$$

$$\int (2x^2 \cos(x^3)) dx$$

$$\text{Let } u = x^3$$

$$\frac{du}{dx} = 3x^2$$

$$du = 3x^2 dx$$

$$\frac{du}{3x^2} = dx$$

$$\int (2x^2 \cos(u)) \frac{du}{3x^2}$$

$$= \frac{2}{3} \int \cos(u) du$$

$$= \frac{2}{3} \sin(u) + c$$

$$= \frac{2}{3} \sin(x^3) + c$$

$$2.3b)$$

$$\int \left(\frac{2x}{\sqrt{1-6x}} \right) dx \text{ (Tutor, 2021)}$$

$$= \int (2x(\sqrt{1-6x})^{-1}) dx$$

$$= \int (2x(1-6x)^{-\frac{1}{2}}) dx$$

$$\text{Let } u = 1-6x$$

$$\frac{du}{dx} = -6$$

$$du = -6dx$$

$$\frac{du}{-6} = dx$$

$$x = -\frac{u-1}{6}$$

$$\int (2(-\frac{u-1}{6})(u)^{-\frac{1}{2}}) \frac{du}{-6}$$

$$= \int (-2(u-1)(u)^{-\frac{1}{2}}) du$$

$$= -2 \int ((u-1)(u)^{-\frac{1}{2}}) du$$

$$= -2 \int (u^{-\frac{3}{2}} - u^{-\frac{1}{2}}) du$$

$$= -2(u^{-\frac{1}{2}} - u^{\frac{1}{2}} + c)$$

$$= -2\left(\frac{1}{\sqrt{u}} - \sqrt{u} + c\right)$$

$$= -2\left(\frac{1}{\sqrt{1-6x}} - \sqrt{1-6x}\right) + c$$

$$2.4)$$

$$I = \int \left(\frac{21x^2}{x^3-x^2-12x} \right) dx$$

$$= \int \left(\frac{21x^2}{x(x^2-x-12)} \right) dx$$

$$= \int \left(\frac{21x^2}{x(x-4)(x+3)} \right) dx$$

$$p(x) : \frac{21x^2}{x(x-4)(x+3)} = \frac{A1}{x} + \frac{A2}{x-4} + \frac{A3}{x+3}$$

$$21x^2 = A1((x-4)(x+3)) + A2(x(x+3)) + A3(x(x-4))$$

Let $x = 0$

$$21(0)^2 = A1((0-4)(0+3)) + A2(0(0+3)) + A3(0(0-4))$$

$$0 = A1((-4)(+3)) + A2(0(3)) + A3(0(-4))$$

$$0 = A1(-12) + 0 + 0$$

$$A1 = 0$$

Let $x = 4$

$$21(4)^2 = A1((4-4)(4+3)) + A2(4(4+3)) + A3(4(4-4))$$

$$336 = A1((0)(1)) + A2(4(7)) + A3(4(0))$$

$$336 = 0 + A2(28) + 0$$

$$A2 = 12$$

Let $x = -3$

$$21(-3)^2 = A1((-3-4)(-3+3)) + A2(-3(-3+3)) + A3(-3(-3-4))$$

$$189 = A1((-7)(0)) + A2(-3(0)) + A3(-3(-7))$$

$$189 = 0 + 0 + A3(21)$$

$$A3 = 9$$

$$p(x) : \frac{0}{x} + \frac{12}{x-4} + \frac{9}{x+3}$$

$$\text{But } I = \int p(x) dx$$

$$= \int \frac{0}{x} dx + \int \frac{12}{x-4} dx + \int \frac{9}{x+3} dx$$

$$= 0 + 12 \int \frac{1}{x-4} dx + 9 \int \frac{1}{x+3} dx$$

$$= 12 \ln(x-4) + 9 \ln(x+3) + c$$

Question 3

3.1)

$$x = 2y \text{ and } x = y^3 - y$$

$$2y = y^3 - y$$

$$y^3 - 3y = 0$$

$$y(y^2 - 3) = 0$$

$$y^2 = 3 \text{ or } y = 0$$

$$y = \sqrt{3}$$

$$\text{Area} = \int_c^d (f(y) - g(y)) dy$$

$$= \int_0^{\sqrt{3}} (2y - (y^3 - y)) dy$$

$$\begin{aligned}
&= \int_0^{\sqrt{3}} (3y - y^3) dy \\
&= \left. \frac{3y^2}{2} - \frac{y^4}{4} \right|_0^{\sqrt{3}} \\
&= \frac{3(\sqrt{3})^2}{2} - \frac{(\sqrt{3})^4}{4} - \left(\frac{3(0)^2}{2} - \frac{0^4}{4} \right) = \frac{9}{4} \text{ units}^2
\end{aligned}$$

3.2)

$$\begin{aligned}
\text{Area} &= \int_a^b (f(x) - g(x)) dx \\
&= \int_1^2 \left(e^x - \frac{1}{x} \right) dx \\
&= \left. e^x - \ln(x) \right|_1^2 \\
&= e^2 - \ln(2) - (e^1 - \ln(1)) = 3.98 \text{ units}^2
\end{aligned}$$

Question 4

4.1a)

$$\begin{aligned}
V &= \int_c^d (2\pi R(y)h(y)) dy \quad (\text{Cao, 2021}) \\
&= 2\pi \int_0^{\frac{\pi}{2}} (y \sqrt{2 \sin^2(y)}) dy \\
&= 2\pi \cdot \sqrt{2} \int_0^{\frac{\pi}{2}} (y \sqrt{\sin^2(y)}) dy \\
&= 2\sqrt{2}\pi \int_0^{\frac{\pi}{2}} (y \sin(y)) dy
\end{aligned}$$

$$\text{Let } I = \int (y \sin(y)) dy$$

$$\text{Let } u = y$$

$$\frac{du}{dy} = 1$$

$$du = dy$$

$$\int v = \int \sin(y)$$

$$v = -\cos(y)$$

$$\begin{aligned}
I &= uv - \int v du \\
&= -y \cos(y) - \int -1 \cos(y) dy \\
&= -y \cos(y) - (-\sin(y)) \\
&= -y \cos(y) + \sin(y) \\
&= \sin(y) - y \cos(y)
\end{aligned}$$

$$\begin{aligned}
V &= 2\sqrt{2}\pi [\sin(y) - y \cos(y)]_0^{\frac{\pi}{2}} \\
&= 2\sqrt{2}\pi \left(\sin\left(\frac{\pi}{2}\right) - \frac{\pi}{2} \cos\left(\frac{\pi}{2}\right) - (\sin(0) - 0 \cos(0)) \right) \\
&= 2\sqrt{2}\pi (1 - 0) = 2\sqrt{2}\pi \text{ units}^3
\end{aligned}$$

4.1b)

Method of cylindrical shells

4.2a)

$$y^{\frac{1}{2}} + x^{\frac{1}{2}} = a^{\frac{1}{2}}$$

$$0^{\frac{1}{2}} + x^{\frac{1}{2}} = a^{\frac{1}{2}}$$

$$x = a$$

$$x = 0$$

$$y = (a^{\frac{1}{2}} - x^{\frac{1}{2}})^2$$

$$f(x) = (a^{\frac{1}{2}} - x^{\frac{1}{2}})^2$$

$$\begin{aligned} V &= \int_a^b \pi(f(x))^2 dx \\ &= \pi \int_0^a ((a^{\frac{1}{2}} - x^{\frac{1}{2}})^2)^2 dx \\ &= \pi \int_0^a ((a^{\frac{1}{2}} - x^{\frac{1}{2}})(a^{\frac{1}{2}} - x^{\frac{1}{2}}))^2 dx \\ &= \pi \int_0^a (a - 2(ax)^{\frac{1}{2}} + x)^2 dx \\ &= \pi \int_0^a ((a - 2(ax)^{\frac{1}{2}} + x)(a - 2(ax)^{\frac{1}{2}} + x)) dx \\ &= \pi \int_0^a (a^2 - 2a(ax)^{\frac{1}{2}} + ax - 2a(ax)^{\frac{1}{2}} + 4ax - 2x(ax)^{\frac{1}{2}} + ax - 2x(ax)^{\frac{1}{2}} + x^2) dx \\ &= \pi \int_0^a (a^2 - 4a(ax)^{\frac{1}{2}} + 6ax - 4x(ax)^{\frac{1}{2}} + x^2) dx \\ &= \pi \left[a^2x - 4a\left(\frac{(a)^{\frac{1}{2}}(x)^{\frac{3}{2}}}{\frac{3}{2}}\right) + \frac{6ax^2}{2} - 4x\left(\frac{(a)^{\frac{1}{2}}(x)^{\frac{3}{2}}}{\frac{3}{2}}\right) + \frac{x^3}{3} \right]_0^a \\ &= \pi \left[a^2x - 4\left(\frac{(a)^{\frac{1}{2}}(x)^{\frac{3}{2}}}{\frac{3}{2}}\right)(a+x) + 3ax^2 + \frac{x^3}{3} \right]_0^a \\ &= \pi(a^2(a) - 4\left(\frac{(a)^{\frac{1}{2}}(a)^{\frac{3}{2}}}{\frac{3}{2}}\right)(a+a) + 3a(a)^2 + \frac{1}{3}(a)^3 - (a^2(0) - 4\left(\frac{(a)^{\frac{1}{2}}(0)^{\frac{3}{2}}}{\frac{3}{2}}\right)(a+0) + 3a(0)^2 + \frac{1}{3}(0)^3)) \\ &= \pi(a^3 - 4\left(\frac{a^2}{\frac{3}{2}}\right)(2a) + 3a^3 + \frac{1}{3}a^3 - 0) \\ &= \pi(a^3 - \frac{16}{3}a^3 + 3a^3 + \frac{1}{3}a^3) \\ &= \pi(-1a^3) = -\pi a^3 \text{ units}^3 \end{aligned}$$

4.2b)

Disk method

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